



Technical Documentation

Minimal dm-Verity Bootloader

Team Oriented Project

Group A

Linux Kernel Extension for DM-Verity Authentication, Verification and Setup

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Cooperation with:

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1. Introduction

1.1. Purpose

The `qemu_main.sh` script serves as a minimal, non-interactive bootloader and development launcher for the dm-verity enabled system. Unlike the `launch_qemu.sh` testing suite, this script's sole purpose is to quickly and reliably boot the original, working root filesystem image (`rootfs.img`) for development, focusing exclusively on the successful kernel-side verification workflow.

Its design is strictly focused on resolving paths and starting QEMU with the exact kernel command line required for the dm-verity autoboot module.

1.2. Architecture Role

This script demonstrates that all critical security logic, including reading the PKCS#7 signature, validating the certificate against the trusted keyring, building the dm-verity mapping, and mounting the verified root - occurs entirely within the kernel (specifically, the dm-verity-autoboot module). The userspace bootloader only handles file location and QEMU parameter passing.

2. Script Specification: qemu_main.sh

2.1. Disk Image Resolution

The script uses the standard Linux command `realpath` to resolve the absolute path of the disk image (`IMG_ABS`) from its relative location (`IMG_REL`). This ensures QEMU can locate the file regardless of the working directory from which the script is executed.

```
22  # Resolve the absolute path for the disk image
23  if ! IMG_ABS=$(realpath "$IMG_REL"); then
24      echo "realpath($IMG_REL) failed" >&2
25      exit 1
26  fi
```

2.2. Kernel Command Line Parameters

The script constructs the kernel command line (`APPEND_CMDLINE`) to delegate the disk setup and mounting process entirely to the kernel's dm-verity module.

```
31  # Kernel command line
32  APPEND_CMDLINE="console=ttyS0,115200 \
33  loglevel=7 \
34  dm_verity_autoboot.autoboot_device=/dev/vda \
35  root=/dev/dm-0 rootfstype=ext4 rootwait rootdelay=10"
```

- `dm\_verity\_autoboot.autoboot\_device=/dev/vda`: Instructs the kernel module to read the dm-verity metadata from the disk attached as `vda`.

- `root=/dev/dm-0`: The kernel attempts to mount the root filesystem from the device mapper target, which will be created by the dm-verity module upon successful verification.

2.3. QEMU Execution Command

The script launches QEMU using detailed, high-security configuration flags suitable for kernel debugging:

- `-m 1024`: Allocates 1024MB of memory.
- `-nodefaults`: Prevents QEMU from creating default peripherals, ensuring a clean environment.
- `-serial mon:stdio`: Directs the console and monitor output to the standard input/output, crucial for debugging.
- `-d guest_errors`: Enables detailed debugging of guest exceptions.

The disk is attached as a raw, whole disk using two arguments:

1. `-drive`: Defines the file path and format.

```
28 # QEMU -drive option: attach the image as a raw, whole disk (/dev/vda in guest)
29 DRIVE_OPT="if=none,id=drv0,format=raw,media=disk,file=${IMG_ABS}"
```

2. `-device`: Attaches the defined drive to the guest as a virtio-blk device, making it visible as `/dev/vda`.

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